```
#include <dht.h>
dht DHT;
#define DHT11_PIN A1

float temperature,humidity;
float tem,hum;

unsigned int count = 1;        //For times count

void setup()
{
   Serial.begin(9600);
}

void loop()
{
      temperature = DHT.read11(DHT11_PIN);    //Temperature detection
      tem = DHT.temperature*1.0;
      humidity = DHT.read11(DHT11_PIN);
      hum = DHT.humidity*1.0;

      DHT.read11(DHT11_PIN);

      Serial.print(F("############    "));
      Serial.print(F("NO."));
      Serial.print(count);
      Serial.println(F("    ############"));
      Serial.println(F("The temperature and humidity :"));
```

[redacted]

```
      Serial.print(F("["));
      Serial.print(DHT.temperature);
      Serial.print(F("°C"));
      Serial.print(F(","));
      Serial.print(DHT.humidity);
      Serial.print(F("%"));
      Serial.print(F("]"));
      Serial.println("");
delay(10000);
      count++;

}
```

```
#define RELAY  10
void setup()
{
Serial.begin(9600);
   pinMode(RELAY, OUTPUT);
}
 void loop()
{
  digitalWrite(RELAY,0);          // Turns ON Relays 1
  Serial.println("Light ON");
  delay(2000);                              // Wait 2 seconds

  digitalWrite(RELAY,1);          // Turns Relay Off
  Serial.println("Light OFF");
  delay(2000);
}
```

Program:

```
int const trigPin = 10;
int const echoPin = 9;
int const buzzPin = 2;
void setup()
{
pinMode(trigPin, OUTPUT); // trig pin will have pulses output
pinMode(echoPin, INPUT); // echo pin should be input to get pulse width
pinMode(buzzPin, OUTPUT); // buzz pin is output to control buzzering
}
void loop()
{
// Duration will be the input pulse width and distance will be the distance to the obstacle in
centimeters
int duration, distance;
// Output pulse with 1ms width on trigPin
digitalWrite(trigPin, HIGH);
delay(1);
digitalWrite(trigPin, LOW);
// Measure the pulse input in echo pin
duration = pulseIn(echoPin, HIGH);
// Distance is half the duration devided by 29.1 (from datasheet)
distance = (duration/2) / 29.1;
// if distance less than 0.5 meter and more than 0 (0 or less means over range)
if (distance <= 50 && distance >= 0) {
// Buzz
digitalWrite(buzzPin, HIGH);
} else {
// Don't buzz
```



```
digitalWrite(buzzPin, LOW);
}
// Waiting 60 ms won't hurt any one
delay(60);
}
```

Program:

```
#include <dht.h>
dht DHT;
#define DHT11_PIN A0
#define RELAY 8

float temperature,humidity;
float tem,hum;

unsigned int count = 1;        //For times count
void setup()
{
  Serial.begin(9600);
pinMode(RELAY, OUTPUT);
}

void loop()
{
      temperature = DHT.read11(DHT11_PIN);    //Temperature detection
      tem = DHT.temperature*1.0;
      humidity = DHT.read11(DHT11_PIN);
      hum = DHT.humidity*1.0;

      DHT.read11(DHT11_PIN);

      Serial.print(F("###########    "));
      Serial.print(F("NO."));
      Serial.print(count);
      Serial.println(F("    ###########"));
      Serial.println(F("The temperature and humidity :"));
      Serial.print(F("["));
      Serial.print(DHT.temperature);
```

[redacted]

```
      Serial.print(F("°C"));
      Serial.print(F(","));
      Serial.print(DHT.humidity);
      Serial.print(F("%"));
      Serial.print(F("]"));
      Serial.println("");
if (tem >= 23   )
 {
digitalWrite(RELAY,0);
}
else
{
digitalWrite(RELAY,1);
}
delay(10000);
      count++;

}
```

Program:

```
// IR Sensor with Arduino
//Digital code
int val = 0 ;
 void setup()
 {
 Serial.begin(9600); // sensor buart rate
 pinMode(7,INPUT);  // IR sensor output pin connected
 pinMode(9,OUTPUT);  // LED
 pinMode(8,OUTPUT);  // BUZZER
 }
 void loop()
```

[redacted]

```
 {
 val = digitalRead(2);  //  IR sensor output pin connected
 Serial.println(val);  // see the value in serial monitor in Arduino IDE
 delay(500);
 if(val == 1 )
 {
 digitalWrite(9,HIGH);  // LED ON
 digitalWrite(8,HIGH);  // BUZZER ON
 }
 else
 {
 digitalWrite(9,LOW);  // LED OFF
 digitalWrite(8,LOW);  // BUZZER OFF
 }
 }
```

Analog program

```
// IR Sensor with Arduino
void setup()
{
Serial.begin(9600); // sensor baud rate
pinMode(9,OUTPUT);  // LED
pinMode(8,OUTPUT);  // BUZZER
}
void loop()
{
int s1=analogRead(A0); //ANALOG PIN FOR SENSOR
Serial.println(s1);   // see the value in serial monitor in Arduino IDE
delay(100);
if(s1>500)
{
digitalWrite(9,HIGH);  // LED ON
digitalWrite(8,HIGH);  // BUZZER ON
}
else
{
digitalWrite(9,LOW);  // LED OFF
digitalWrite(8,LOW);  // BUZZER OFF
}
}
```

Program:

```
// LDR Sensor with Arduino
//Digital code
int val = 0 ;
 void setup()
 {
```

[redacted]

```
 Serial.begin(9600); // sensor buart rate
 pinMode(7,INPUT);  // IR sensor output pin connected
 pinMode(3,OUTPUT);  // LED
 }
 void loop()
 {
 val = digitalRead(7);  //  IR sensor output pin connected
 Serial.println(val);  // see the value in serial monitor in Arduino IDE
 delay(500);
 if(val == 1 )
 {
 digitalWrite(3,HIGH);  // LED ON

 }
 else
 {
 digitalWrite(3,LOW);  // LED OFF
}
 }
```

Analog program

```
// IR Sensor with Arduino
void setup()
{
Serial.begin(9600); // sensor baud rate
pinMode(3,OUTPUT);  // LED

}
void loop()
{
int s1=analogRead(A0); //ANALOG PIN FOR SENSOR
Serial.println(s1);   // see the value in serial monitor in Arduino IDE
delay(100);
if(s1>200)
{
digitalWrite(3,HIGH);  // LED ON

}
else
{
digitalWrite(3,LOW);  // LED OFF

}
}
```

Results and Conclusion:

Program:

```
% create an arduino object

a = arduino('com19', 'uno');

% start the loop to blink led for 10 seconds

for i = 1:10

writeDigitalPin(a, 'D11', 1);
writeDigitalPin(a, 'D12', 0);


pause(0.5);

writeDigitalPin(a, 'D11', 0);
writeDigitalPin(a, 'D12', 1);


pause(0.5);

end

% end communication with arduino

clear a
```

```
#include <dht.h>
dht DHT;
#define DHT11_PIN A1

float temperature,humidity;
float tem,hum;

unsigned int count = 1;        //For times count

void setup()
{
   Serial.begin(9600);

[redacted]

}

void loop()
{
      temperature = DHT.read11(DHT11_PIN);    //Temperature detection
      tem = DHT.temperature*1.0;
      humidity = DHT.read11(DHT11_PIN);
      hum = DHT.humidity*1.0;

      DHT.read11(DHT11_PIN);

      Serial.print(F("###########    "));
      Serial.print(F("NO."));
      Serial.print(count);
      Serial.println(F("    ###########"));
      Serial.println(F("The temperature and humidity :"));
      Serial.print(F("["));
      Serial.print(DHT.temperature);
      Serial.print(F("°C"));
      Serial.print(F(","));
      Serial.print(DHT.humidity);
      Serial.print(F("%"));
      Serial.print(F("]"));
      Serial.println("");
delay(10000);
      count++;

}
```

MATLAB DATA ACQUISITION CODE

```
s = serial('COM18');
time=100;
i=1;
while(i<time)
fopen(s)
fprintf(s, 'Your serial data goes here')
out = fscanf(s)
Temp(i)=str2num(out(1:4));
 subplot(211);
 plot(Temp,'g');
 axis([0,time,20,50]);
title('Parameter: DHT11 Temperature');
xlabel('---> time in x*0.02 sec');
ylabel('---> Temperature');
grid
Humi(i)=str2num(out(5:9));
 subplot(212);
 plot(Humi,'m');
```

[redacted]

```
axis([0,time,25,100]);
title('Parameter: DHT11 Humidity');
xlabel('---> time in x*0.02 sec');
ylabel('---> % of Humidity ');
grid
fclose(s)
i=i+1;
drawnow;
end
delete(s)
clear s
```